Mohammadali Sefidi Esfahani

PhD Student in Computer Science at Sharif University of Technology

Google Scholar | Homepage | Linkedin | Github | Email

Education

PhD in Computer Science — *Sept. 2026 - Present*

Sharif University of Technology — *Iran, Tehran*

- **Supervisors:** Dr. Seyed Reza Moghadasi & Dr. Fatemeh Seyyedsalehi
- *Ranked 4th in the national PhD entrance exam*

PhD in Software Engineering (Withdraw) — *Sept. 2024 - Sept. 2026*

Concordia University — *Montreal, Canada*

- **Research**: SE Knowledge Management Systems
- **Supervisor:** Professor Peter C. Rigby
- **Withdraw due to not receiving visa**

Master of Science in Computer Science (Soft Computing and Artificial Intelligence) — *Sept. 2021 - Sept. 2023*

Amirkabir University of Technology — *Tehran, Iran*

- **Cumulative GPA:** 19.62 out of 20
- **Master's Thesis:** Identification of Unspecified Events from Twitter
- **Supervisor:** Dr. Mohammad Akbari
- *Graduated as a 1st-ranked student*

Bachelor of Science in Computer Science — *Sept. 2017 - July 2021*

Kharazmi University — *Tehran, Iran*

- **Cumulative GPA:** 18.26 out of 20
- **Bachelor's Thesis:** Comparative Study on the Performance of ML Models for Fall Detection in Older People
- **Supervisor:** Dr. Mohammad Soltanian
- *Graduated as a 1st-ranked student*

Honors and Awards

Year	Award / Institution	Value
2026	**Canada Impact+ Research Training Award** Natural Sciences and Engineering Research Council (NSERC), Canada	CA$120,000
2026	**Tuition Award of Excellence** Concordia University, Canada	CA$48,888
2025	**Graduate Doctoral Fellowship** Concordia University, Canada	CA$56,000
2024	**Split Graduate Fellowship GCS** Concordia University, Canada	CA$28,000
2023	**Ranked 1st among the MSc. computer science students (2021 entrance)** Amirkabir University of Technology, Iran	
2021	**Ranked 1st among the BSc. computer science students (2017 entrance)** Kharazmi University of Technology, Iran	

Publications

Mitigating the Risk of Defects and Improving Knowledge Distribution with Code Reviewer Recommenders

Mohammadali Sefidi Esfahani & Peter C. Rigby, International Conference on the Foundations of Software Engineering (FSE'26), Montreal, Canada, 2026

- **Link:** ACM

The Cost vs the Benefit of Adding an Extra Code Reviewer to Mitigate Developer Turnover through Reviewer Recommenders

Mohammadali Sefidi Esfahani & Fahimeh Hajari & Peter C. Rigby, International Conference on Software Engineering (ICSE'26), Rio, Brazil, 2026

- **Link:** ACM

Can Perplexity Serve as a Cognitive Signal for Code Understandability?
Xiaokai Rong & Mohammadali Sefidi Esfahani & Aashish Yadavally & Peter C. Rigby & Tien N. Nguyen, **Submitted to ICSE'27**, 2026
- **Link:** Arxiv

Allocating Human Review Effort from Low-Risk to Defect-Prone Pull Requests with LLM Supplementary Knowledge
Mohammadali Sefidi Esfahani & Peter C. Rigby, **Submitted to FSE'27**, 2026
- **Link:** Under Review

EnrichEvent: Enriching Social Data with Contextual Information for Emerging Event Extraction
Mohammadali Sefidi Esfahani & Mohammad Akbari, Iran Journal of Computer Science, 2025
- **Link:** Springer

Comparative Study on Performance of ML Models for Fall Detection in Older People
Mohammadali Sefidi Esfahani & Mohammad Fattahian, Journal of Artificial Intelligence, Virtual Reality, and Human-Centered Computing, 2025
- **Link:** AI-VR-HCC Journal

Feature Learning for Identification of Persian Events from Twitter
Mohammadali Sefidi Esfahani & Mohammad Akbari, CSICC'23, 2023
- **Link:** Civilica & PDF

Experience

Graduate Research Associate at Concordia University *Sept. 2024 - Sept. 2026*
Supervisor: Professor Peter C. Rigby
- **Duties:** Research on SE knowledge management systems and the impact of developer turnover.

Graduate Research Assistant in Data Science Innovation Center *Nov. 2021 - Sept. 2024*
Supervisor: Dr. Mohammad Akbari
- **Duties:** Research, Management of Data Science Innovation Center weekly meetings, Mentoring undergraduate members

Teaching Assistant (Machine Learning) *Sept. 2023 - Feb. 2024*
Lecturer: Dr. Ali Mohades Khorasani
- **Duties:** Programming assignments, Grades of programming assignments, Troubleshooting sessions

Lead Teaching Assistant (Machine Learning) *Sept. 2022 - Feb. 2023*
Lecturer: Dr. Mohammad Akbari
- **Duties:** Assignments, Grades of assignments and final project, Troubleshooting sessions

Graduate Research Assistant in Natural Language Processing Innovation Center (NLPIC) *Nov. 2021 - Aug. 2022*
Supervisors: Dr. Ali Mohades & Dr. Mohammad Akbari & Dr. Mohammad Ebrahim Shiri
- **Duties:** Research

Member of ACM association *March 2021 - Aug. 2021*
Supervisor: Dr. Bardia Panahbehagh
- **Duties:** Event planing and management

Projects

Sofia-ExpertPlusPlus (GitHub)
Replication Package for FSE'26 Paper
- In this paper, we design reviewer recommendation strategies to mitigate the risk of defects and improve knowledge distribution.

Sofia-LearnerPlus (GitHub)
Replication Package for ICSE'26 Paper
- In this paper, we investigate the cost vs. benefit of assigning an extra reviewer to pull requests with knowledge at risk to turnover.

EnrichEvent (GitHub)
Replication Package for my Master Thesis
- An end-to-end framework for identifying unknown events from Twitter in an unsupervised setting.

Comparative Study of ML Models for Fall Detection in Older People (GitHub)
Replication Package for my Bachelor Thesis
- This project aimed to implement and evaluate the efficiency of diverse ML models for fall detection in older people.

No-Show Appointments Prediction

Real-World Project (Private-Access)

- This project aimed to implement an end-to-end framework for no-show appointment prediction based on patient history.

Data Generator for Comparative Opinion Summarization using Chat-GPT (GitHub)

Data Science Innovation Center Project

- The implemented tool scrapes users' reviews for hotels from Eghamat24.com website and builds a synthetic dataset for comparative opinion summarization.

Small Projects in AI (GitHub) / Small GNN-based Projects (GitHub)

Self-Study

- Collection of small projects in various fields like ML, Data Mining, Deep Learning, NLP, and GNNs

Research Interest

- Trustworthy AI
- Explainable AI
- Applied ML for Industrial Systems
- Code Reviewer Recommendation
- AI-assisted Code Review
- AI and ML for Software Engineering

Skills

Programming Languages	Python, Java, C-sharp
Libraries and Frameworks	TensorFlow, Pytorch, Pytorch-Geometric, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, dotnet
Tools	Git, Latex, Adobe Photoshop, Search Engine Optimization (SEO)

Languages

English	MSRT (Overall: 75/ Listening: 25/ Structure & Written Expression: 21/ Reading Comprehension: 29/ Test Date: 26 September 2025)
English	TOEFL iBT (Overall: 98/ Reading: 24/ Listening: 26/ Speaking: 22/ Writing: 26/ Test Date: 18 November 2023)

Certificates

Deep Learning Specialization (Credential link) *Nov. 2023*

Issued by Coursera: DeepLearning.AI

- Neural Networks and Deep Learning
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Structuring Machine Learning Projects
- Convolutional Neural Networks
- Sequence Models

Natural Language Processing Specialization (Credential link) *Nov. 2023*

Issued by Coursera: DeepLearning.AI

- Natural Language Processing with Classification and Vector Spaces
- Natural Language Processing with Probabilistic Models
- Natural Language Processing with Sequence Models
- Natural Language Processing with Attention Models

Machine Learning Specialization (Credential link) *Oct. 2023*

Issued by Coursera: DeepLearning.AI and Standford

- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Unsupervised Learning, Recommenders, Reinforcement Learning

Task-oriented course in artificial intelligence and machine learning (Credential link) *Nov. 2021*

Issued by Quera